

Deriving Fama and French Factors

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Abstract. Fama and French (2015) introduce an accounting-based pricing model to justify book-to-price, profitability, and investment as characteristics from which factor mimicking portfolios can be formed. Given skepticism about factors elicited from data mining, the modeling is important. However, the comparative statics in the modeling are inconsistent with how accounting actually works. This paper explains, then introduces an alternative factor construction based on accounting principles that generate the accounting numbers. Empirical tests support the critique with the alternative factor construction comparing favorably to Fama and French factors in explaining stock returns.

Deriving Fama and French Factors

Introduction

For a considerable time, research in finance has been in pursuit of an operational asset pricing model with risk and expected return described in terms of observables. Ideally, the model would be prompted by theory, as with the standard Capital Asset Pricing Model (CAPM) but, in the absence of theory, Fama and French (1992,1993) took the sensible approach of letting the data speak: Observables were gleaned by finding characteristics associated with stock returns in the data, with risk factors then constructed based on these observables. Accounting numbers, like book value, profitability, and investment feature prominently. That, on the face of it, is compelling because risk to equities is in firms' activities and accounting reports on those activities.

Robust correlations are to be given some weight and the consequent factor models fit well in describing asset returns in sample. However, correlations without theory are less persuasive; typically, the econometrician first specifies a model based on theory then takes the model to the data for testing. The proliferation of factors “discovered” from data mining has prompted the “factor zoo” complaint. Attempts to reduce the zoo to a model with some parsimony has only involved further data analysis applying goodness-of-fit criteria, but again with theory absent.¹

Fama and French (2015) respond to this state of affairs with a formal model that expresses price in terms of accounting numbers and the expected return, and then proceed with comparative statics to show how the measures in their extended factor model——book-to-market, profitability (the book rate of return), and investment——convey the expected return. This modelling is the focus of our paper. First, we explain that the comparative statics are inconsistent with how accounting actually works. The expected return is inferred by varying one of the three accounting measures while holding the other two constant. But accounting numbers are codetermined in a double-entry system. The accounting for earnings also determines book value, for example, and thus book-to market, profitability (earnings-to 3-book value) and the

¹ Harvey, Liu, and Zhu (2015), Harvey and Liu (2019), and Green, Hand, and Zhang (2017) point to the dangers of data mining.

investment component of book value; the three are jointly determined. Thus, one cannot vary one while holding the other accounting measures constant. Second, risk is conveyed by how accounting numbers are measured and there is no such prescription in the modeling; random accounting numbers satisfy the modeling equation. The ability of book-value, profitability, and investment to convey risk and return depends on the accounting principles underlying these measures.

Working from the same pricing model with the same accounting numbers but with this understanding, we proceed to identify factors by recognizing how the accounting numbers conveys risk and the expected return under accounting principles. In contrast to the critiqued comparative statics, we explain how accounting numbers, jointly determined, relate to each and jointly convey the expected return. Of particular importance is the recognition that

$\frac{Earnings}{Book\ Value} \times \frac{Book\ Value}{Price} = \frac{Earnings}{Price}$, that is, profitability x book-to-market in the Fama and French models convey earnings-to-price. Thus, the E/P ratio that is implicit in the Fama and French modeling comes into play, sensibly, for it is earnings that are at risk in the accounting-based pricing model. That leads to the question of the measurement of earnings and how E/P conveys risk and return under accounting principles.

The recognition of these accounting features promotes an alternative factor identification. A comparative empirical analysis with the Fama and French factors supports the alternative identification.

A Critique of the Fama and French Modelling

The Fama and French (2015) modelling seeks to justify the book-to-market (BE/ME) “value” factor (HML) identified in the original Fama and French (1993) three-factor model as well as the profitability (RMW) and investment (CMA) factors added in their extended five-factor model. The modelling expresses price in terms of accounting numbers by substituting dividends in the familiar dividend discount model with earnings and book value of equity, B . Assuming clean-surplus accounting,

$$B_t = B_{t-1} + Earnings_t - Net\ Dividends_t \quad (1)$$

and thus $Net\ Dividends_{\tau} = Earnings_{\tau} - (B_{\tau} - B_{\tau-1})$, all τ . Substituting for dividends in the dividend discount model with a long-term discount rate (expected return), r ,

$$P_t = \sum_{\tau=1}^{\infty} [Earnings_{t+\tau} - (B_{t+\tau} - B_{t+\tau-1})] / (1+r)^{\tau}. \quad (2a)$$

Dividing through by B_t ,

$$\frac{P_t}{B_t} = \frac{\sum_{\tau=1}^{\infty} [Earnings_{t+\tau} - (B_{t+\tau} - B_{t+\tau-1})] / (1+r)^{\tau}}{B_t}. \quad (2b)$$

Variables subscripted $\tau > 0$ are expected values given information at time t .² Equation 2b is the expression from which Fama and French (FF) make predictions. The term, $\frac{\sum_{\tau=1}^{\infty} Earnings_{t+\tau}}{B_t}$, the expected book rate of return, is called profitability by FF, with a measure of the current profitability the indicator of future profitability. The change in book value, $\frac{B_{t+\tau} - B_{t+\tau-1}}{B_t}$ is given the label, “growth in investment.” Reverse-engineering the expression, and carrying out some comparative statics connects these measures to the expected return (and the required return if risk is efficiently priced). FF conclude that:

- (i) r is increasing in B/P holding the expected book rate of return and expected investment growth constant,
- (ii) r is increasing in the expected book rate of return holding B/P and expected investment growth constant, and
- (iii) r is decreasing in expected investment growth holding B/P and expected book rate of return constant.

B/P, return on book value (profitability), and investment then become characteristics from which factors are formed.

Before getting into the comparative statics, note the following equivalences:

First, equation 2a can be restated. As $P_t = [Earnings_{t+1} - (B_{t+1} - B_t) + P_{t+1}] / (1+r)$ and iterating out over future $P_{t+\tau}$,

$$P_t = B_t + \sum_{\tau=1}^{\infty} [Earnings_{t+\tau} - r \cdot B_{t+\tau-1}] / (1+r)^{\tau}$$

² Fama and French use the notation ME and BE for the market value and book value of equity. We use the notation for a market-to-book ratio, P/B, that is common in practice.

and

$$\frac{P_t}{B_t} = 1 + \frac{\sum_{\tau=1}^{\infty} [Earnings_{t+\tau} - r \cdot B_{t+\tau-1}] / (1+r)^{\tau}}{B_t}. \quad (2c)$$

This equivalent to equation 2b is the more familiar residual earnings model with P/B given by expected residual earnings relative to current book value. Further, earnings add to book value under the clean surplus relation (1) so, iterating out over future τ adds another equivalent pricing equation,

$$P_t^T = \frac{\sum_{\tau=1}^T [Earnings_{t+\tau} + Dividend_{t+\tau} ((1+r)^{T-\tau} - 1)]}{(1+r)^T - 1} \quad (2d)$$

$\rightarrow P_t$ in equation 2a as $T \rightarrow \infty$ (Ohlson 1995, Appendix 3). The numerator is future cum-dividend earnings where dividends are reinvested on personal account in a risk-free account that can be drawn on for consumption or reinvested in the firm to gain subsequent $Earnings_{t+\tau}$. Reverse engineering this equation,

$$(1+r)^T = \frac{\sum_{\tau=1}^T [Earnings_{t+\tau} + Dividend_{t+\tau} ((1+r)^{T-\tau} - 1)]}{P_t^T} - 1. \quad (2e)$$

Thus, r is given by the expected cum-dividend earnings-to-price (with price in the denominator discounted for the risk to future earnings). This resonates with the idea that it is future earnings that are at risk and price in the denominator discounts for that risk.

Second, the progression from the dividend discount model to equation 2a and all subsequent equations is by substitution, and mere substitution puts nothing on the table. Without specification of how earnings and book values are measured, these equations have little meaning. Indeed, earnings and book value can be random numbers (subject to the standard “transversality” condition on the dividend discount model).³ Accounting principles determine the measurement of earnings and book value so the modeling cannot be substantively applied without an

³ The transversality condition on the dividend discount model requires expected dividends to be ultimately paid out, bringing expected book value to zero. Correspondingly, a transversality condition on equation 2a requires expected book value to go to zero. That enforces the equivalence of price in equation 2a to that in the dividend discount model. However, up to the liquidating dividend that brings book value to zero, earnings and book value in the model tolerate any measurement (with the settling up to dividends in the long run).

understanding of accounting principles, in particular how those principles yield numbers that convey risk and r .

The Comparative Statics⁴

Property (i).

Consider a relatively low-risk money-market fund and a relatively risky equity fund. For both funds, $B/P = 1$, so B/P cannot distinguish risk. This is a property of the accounting applied: Both funds apply mark-to-market (or fair value) accounting to price book value (NAV), and this accounting means B/P cannot discriminate on risk and the required return. In contradiction to property (i), r varies without variation in B/P , and B/P is unchanged irrespective of the conditioning book rate of return and investment.

This fair value accounting case makes the point: One cannot connect accounting numbers to risk and return without specification of accounting measurement that conveys that risk. If B/P is to indicate risk and return, it must be accounting that differs from fair value accounting. While some financial institutions employ fair value accounting, the prevailing accounting under U.S. GAAP and IFRS is historical cost accounting whereby book value is at (amortized) historical cost that is added to only when earnings from investment are realized, as in the clean-surplus relation (1).

However, the comparative statics in (i) that do not work with fair-value accounting do not work with historical cost accounting either, for book value and profitability (return on equity, ROE) are co-determined in the accounting system: a lower book value means a higher rate of return on book value (profitability) *ceteris paribus*. A firm with missing assets on the balance sheet and thus a low B/P (because “intangible assets” like R&D and brand building are not booked to the balance sheet) reports high profitability when the earnings from the missing assets are compared to the low book value. This is clear in equation 2b: Add b_t to book value (or subtract it) such that $\text{price-to-book} = \frac{P_t}{B_t + b_t}$ then the forward book rate of return, $\frac{Earnings_{t+1}}{B_t + b_t}$ correspondingly changes. One cannot hold book rate of return constant in (i) while varying B/P ; they are codetermined by

⁴ Parts of this critique are the point of departure for the the accounting-based pricing in Penman, Reggiani, Richardson, and Tuna (2018) and (with reference to this paper) in Penman and Zhang (2024).

the accounting. It is also clear from equation 2d where price is based on capitalized expected earnings. Dividing through by B_t states P/B in terms of the expected book rate or return, but with no change in price or r . This is just due to the measurement of book value affecting both.

The Coca Cola Company (KO) traded at a market capitalization of \$263.05 billion on April 1, 2024, on equity book value of \$25.94 billion (a P/B of 10.1), due in part from its missing brand asset on the balance sheet. With the earnings from the brand coming through the income statement, the analysts' consensus forward earnings forecast at the time was \$12.11 billion or a return on equity (ROE) of 46.7%. Suppose the brand asset, estimated at \$60 billion (say) were booked to the balance sheet, resulting in book equity of \$85.94 billion and P/B of 3.1. That accounting also reduces the forward return on equity to 14.1. Both P/B and the book return are determined by the accounting for assets.⁵

As $Book\ Value_t = Assets_t - Liabilities_t$, this accounting for assets affects equity book value. But another feature means the return on equity and B/P are codetermined: Stock repurchases financed by borrowing (or debt-for-equity swaps) reduce book value without any effect on assets producing earnings, thus typically increasing the book rate of return.⁶ The increasing number of companies with negative book value because of stock repurchases but with demonstrated profitability (McDonalds, Starbucks, Yum! Brands, Papa John's, Denny's) are exhibits for the point. For the typical $B/P < 1.0$, B/P decreases with this financing and increases for $B/P > 1.0$. As with the accounting for assets, this is so for all profitability measures denominated in book value, including the operating profitability measure in the FF five-factor and six-factor models.

The codetermination of profitability and B/P is evident with the correlation of B/P and book return in Fama and French (2000). Panel A of Table 1 updates to 2022. B/P is grouped into low, medium, and high portfolios from the cross section each year, with the mean ROE for these B/P portfolios then reported. ROE is negatively correlated with B/P for positive ROE (ROE^+). When negative ROE (loss firms) are included, the correlation is diminished as loss firms are associated

⁵The added book value has to be amortized against earnings but with no effect on earnings if the current expenditure on advertising and promotion that maintains the brand asset equals the amortization expense.

⁶ The financing also reduces earnings in the numerator in the book rate of return due to higher interest expense from the issued debt, but the proportional effect on the denominator is typically greater than that on the numerator.

with all levels of B/P, including both extremes of B/P. So, there is not much variation on B/P holding negative ROE constant. The mean cross-sectional Pearson (Spearman) correlation between B/P and (all) ROE over years is -0.118 (-0.317) and -0.365 (-0.589) for positive ROE.

Property (ii).

In similar vein, one cannot vary the book rate of return while holding B/P constant. But another accounting feature enters. Assets are missing from book value because (intangible) investments have been charged against earnings. So, while this accounting reduces the denominator in the book rate of return, increasing the measure, it also reduces the numerator; a profitability measure is the product of the accounting that affects both the numerator and denominator of the book rate of return. For this additional reason, one cannot vary ROE while holding book value and B/P constant, for the accounting for book value again affects ROE.

Property (iii).

$\frac{B_{\tau} - B_{\tau-1}}{B_t}$ is labelled as growth in investment. But investments in a business are made by

reducing the cash component of assets and increasing the investment component (debit and credit) with no effect on total assets or B_t , nor B_{τ} in expectation. And added investment financed by debt increases $Assets_{\tau}$ and $Liabilities_{\tau}$ by the same amount with no effect on B_t .

Further, $\frac{B_{t+\tau} - B_{t+\tau-1}}{B_t} = \frac{Earnings_{t+\tau} - Net Dividends_{t+\tau}}{B_t}$ by the clean-surplus relation in equation 1, so incorporates the book rate of return $\frac{Earnings_{t+\tau}}{B_t}$: One cannot hold the expected book rate of return constant while changing $\frac{B_{t+\tau} - B_{t+\tau-1}}{B_t}$. Similarly, $\frac{B_{t+\tau} - B_{t+\tau-1}}{B_t}$ is reduced by net dividends which is a financing transaction not necessarily related to investment (and, indeed, independent of investment in Miller and Modigliani (1961) which also invokes equation 2a).

E/P in the Fama and French Analysis

Equation 2d states the FF equation 2a in terms of expected earnings-to-price. Trailing earnings-to-price (E/P) and forward (forecasted) earnings-to-price ratios are strongly correlated, a property of earnings, as measured, that is quite different from that of random numbers permitted in the FF

model as it stands.⁷ As the trailing and forward E/P are denominated in the same price that discounts for risk, observed trailing E/P is a contender as an indicator of the expected return in equation 2e.

In the seminal FF 1992 paper which led to the 1993 three-factor model, E/P was entertained but was said to be subsumed by B/P. And well it might as E/P and B/P are positively correlated. Panel B of Table 1 reports that, for positive E/P (E^+/P), higher B/P is associated with higher E/P, though loss firms are again associated with both high and low B/P. The mean cross-sectional Pearson (Spearman) correlation between B/P and (all) E/P over years is 0.118 (0.209) and 0.498 (0.446) with only positive E/P (E^+/P). Given equation 2d, that gives credibility to B/P as an indicator of the required return. However, in Panel C of Table 1, five E^+/P portfolios are formed each year within each B/P portfolio. The positive correlation between B/P and E/P numbers is evident on the diagonal from (Low, Low) to (High, High), but there are numbers off the diagonal: High (low) E/P can be associated with low (high) B/P. And that is so with loss firms included in E/P in Panel D.⁸

However, the FF analysis implicitly incorporates E/P.

Prompted by their comparative statics, FF add profitability to B/P in the five-factor model. In going to the data, a positive correlation of profitability with stock returns is observed, with high profitability then deemed higher risk. However, in cross-sectional regressions the positive correlation on profitability is conditional on B/P in the regression. But,

$$\frac{Earnings_{t+1}}{B_t} \times \frac{B_t}{P_t} = \frac{Earnings_{t+1}}{P_t}. \quad (3)$$

That is, adding the forward book rate of return to B/P yields the forward E/P (and similarly so for the trailing E/P). The E/P here is recovered multiplicatively but does so additively in regressions with logged variables that are typically run. In contrast to property (ii) in the comparative statics,

⁷ Ball and Watts (1972) reported that earnings typically follow a submartingale process, implying that current earnings (plus a drift) is a good indicator of future earnings, a finding that still holds.

⁸ Penman and Reggiani (2018) show that, for a given E/P, it is high B/P rather than low B/P that indicates growth, and that growth is of higher risk of not being realized.

varying ROE while holding B/P constant yields E/P. Book value which codetermines B/P and ROE and thus frustrates the comparative statics drops out.⁹

The ability of E/P to predict returns has been documented consistently. Basu (1977, 1983), in a challenge to the CAPM, reported this early on, though the Grahamite fundamentalists of old were very much aware. Ball (1978) conjectured that E/P is a yield with a discount in the denominator price for risk so should predict returns for risk. Equation 2e is more explicit: r is indicated by expected future earnings-to-price, so current earnings to that same price is an indicator of r if it forecasts future earnings to price (as it does). Indeed, if current earnings were a sufficient statistic for forecasting future earnings, price in equation 2d is just current earnings capitalized as a perpetuity and $E/P = r$. This suggests that E/P is a starting point for indicating risk and return rather than B/P.¹⁰

However, E/P is presumably not generally a sufficient indicator of future earnings-to-price, so further information is needed to satisfy equation 2e.

Method in the Madness

The idea of connecting profitability to risk and the expected return seems to come from the interpretation of the measure as (“real”) economic profitability. An economic rate of return on investment compares the return coming from the investment (in the numerator) to the investment that generate that return (in the denominator). But the accounting rate of return mixes investment with return from investment in its numerator and leaves investment out of the denominator.

However, there is method in the madness that can be exploited in connecting the accounting numbers to risk and return, as will be seen. The commentary on property (ii) explains that, under accounting principles, earnings are reduced by certain investments like R&D and brand building that are charged against earnings. This upsets E/P as a yield from investment, for now earnings are reduced by investment but that investment potentially projects higher future earnings.

Further, the seemingly mad accounting principles potentially convey risk and a discount to price

⁹ With data fitting to find the best profitability measure for goodness of fit, Fama and French (2015) segway to profitability measures other than ROE. We investigate later whether this finesses the issue.

¹⁰ Interesting, Fisher Black (in Black 1980) advocated for a measurement of earnings that is a sufficient statistic for forecasting future earnings.

in the E/P: Investments are expensed because they are deemed to be riskier than investments booked to the balance sheet. The marquee example is R&D: The probability of payoff is relatively low——85% of R&D is unsuccessful, we read——so it is not booked to the balance sheet with the pretense that it provides collateral.¹¹ But the accounting is applied to advertising to build brands——the customers might not turn up——to investments in human capital——the trained employees might leave and go to the competition——and to start-up costs, customer loyalty programs, investment in supply chains and distribution systems, software, and more.

The accounting is linked to risk under the formal accounting principles of the U.S. Financial Accounting Standards Board (FASB) and the International Accounting Standards Board that promulgates International Accounting Standards (IFRS): Assets are booked to the balance sheet under a “probability of future benefits” criterion whereby investments are not booked to the balance sheet with the pretense of providing collateral if outcomes are relatively uncertain.¹² Inventory and property, plant, and equipment producing revenues are deemed to meet the threshold but not many so-called intangible assets yet to yield revenues. However, this is not just a whim of regulators. Penman and Zhang (2020) show analytically that this so-called conservative accounting conveys information about the discount factor is the general, no-arbitrage asset pricing model and, correspondingly, the Lintner-Fama Ratio that determines the discount rate. Penman (2021) provides a review of empirical research that confirms.

Accordingly, while the E/P ratio comes to the fore in equations 2e and 3, there is this additional feature that disturbs E/P for indicating the expected earnings yield, but which conveys additional information about risk and return.

¹¹ Only 0.02% of drugs that enter preclinical testing make it to market and only 20% to 30% of those cover their cost, according to the *Wall Street Journal*, December 5, 2024, p. A16. A review of R&D success rates by the Geneva Graduate Institute Global Health Centre is at <https://www.knowledgeportalia.org/r-d-time-and-success-rate#:~:text=Phase%20I%20to%20Phase%20II,was%20estimated%20to%20be%2011.83%25>.

¹² For the accounting for R&D, FASB Statement No. 2 requires the investment to be expensed due to the “uncertainty of future benefits.” In IAS 38, the IASB applies the criterion of “probable future economic benefits” to distinguish between “research” (which is expensed) and “development” (which is capitalized in the balance sheet and amortized). Even IAS 16 on property, plant, and equipment requires benefits to be “probable” for the asset to be booked. Under the FASB and IASB Conceptual Framework, an uncertainty criterion is applied for an asset to be booked to the balance sheet. This does not imply that the formal accounting requirements always “get it right” but to demonstrate the application of the principle in practice.

The Fama and French Empirical Analysis: Comparative Tests

These considerations prompt an alternative factor identification to take to the data.

For this investigation, firms' annual accounting data are sourced from Compustat for years, 1964-2022, except for financial firms (SIC codes 6000-6999) and utilities. Firms classified as Membership Organization or Unknown SIC are also excluded. Annual returns are compounded monthly returns from CRSP. The appendix details the calculation of variables used in the empirical analysis.

The data mining approach typically runs cross-sectional “characteristic regressions” to identify features that are correlated with future stock returns. “Factor mimicking” portfolios based on these features are then constructed, as in the Fama and French (1993) factor construction paper that builds from their 1992 characteristic regression paper. We follow this approach to compare the factors coming from the FF comparative-static modelling with those prompted by the critique of that modelling.

Characteristic Regressions

The characteristic regression results are in Table 2. Forward returns, the dependent variable, are annual buy-and-hold compounded monthly returns beginning three months after fiscal-year end by which time the explanatory accounting variables must be published, by law. There is an accommodation for firms not surviving for the full year, as described in the appendix. The reported coefficients are means of those estimated from annual cross-sectional regressions, with t-statistics that mean relative to its standard error estimated from the time series of estimated coefficients (Fama and Macbeth style). The regressions are decile-rank regressions because factor portfolios are typically formed from rankings.¹³

In Regression I, B/P by itself predicts returns, the basis for the HML factor. Regression II adds ROE as a predictor, delivering a positive mean coefficient, supporting the addition of a

¹³ The regressions for each calendar year involve firms with different fiscal-years in that year so there is some overlapping of returns in successive years. However, results are similar with the return period beginning at April 31 each year with firms' last annual report numbers at that date, and with only December 31 firms at that date where the numbers are most recent. Regression results with continuous variables (with an outlier treatment) are similar, as are those with logged variables.

profitability factor with the claim that higher ROE indicates higher risk and expected return. However, ROE added to B/P yields E/P in equation 3 and Regression (III) confirms that E/P is an indicator of returns. That supports the Ball (1978) conjecture that E/P, as a yield, should predict returns and the more explicit formation in equation 2e where r is given by expected earnings relative to the same price in E_t/P_t ; current earnings-to-price is a predictor of future earnings -to-price¹⁴

With the addition of ROE to E/P in Regression (IV), the coefficient on ROE turns negative, and significantly so. This challenges the claim that higher ROE indicates higher risk and questions a profitability factor based on it. But why is the coefficient on ROE negative? The conservative accounting explains. With investment expensed against earnings if it is particularly risky, E/P is affected. But that expensing of investment also reduces earnings in the numerator of ROE; a lower ROE, given E/P, indicates higher risk. Furthermore, if that risky investment pays off with earnings, ROE is high because those earnings are on a low base (the investment is not on the balance sheet), and risk resolved means lower risk. Accordingly, ROE is negatively correlated with returns conditional on E/P in the regression. That agrees with the intuition that firms with higher profitability are lower risk.

This conservative accounting explanation is demonstrated analytically in Penman and Zhang (2020) and Regression (V) further supports it: Adding the risky expensed investment (denominated in the same (risk-discounted) price as in E_t/P_t) reports a positive mean coefficient; Expensed investment conveys higher risk and expected return. The coefficient on ROE is no longer significant.¹⁵ In short, returns are explained by E/P and the extent to which E/P is affected by conservative accounting for investment.¹⁶ The finding promotes current E_t/P_t as the primary

¹⁴ The regressions involve firms with positive earnings. As future earnings yields must be positive (for positive prices), positive earnings yields are a base for projecting future earnings yields. A mean negative coefficient on E/P is observed when running the regressions with loss firms which, when applied to a negative E/P implies positive returns.

¹⁵ Although ROE is the profitability measure in the FF modelling, they embrace an alternative profitability measure, “operating profitability” on the basis of goodness of fit. Operating profitability adds no explanatory power to Regression (V). The mean cross-sectional rank correlation between operating profitability and ROE_t is 0.854.

¹⁶ Several papers, for example, Li (2020), Park (2022), Arnott, Harvey, Kalesnik, and Linnainmaa (2021), Eisfeldt, Kim, and Papanikolaou (2020), and Eisfeldt and Papanikolaou (2013), correct the (seeming madness) of the accounting, by removing the expensed investment to book value. However, that obscures the accounting that

predictor of future earnings denominated in the same P_t as in equation 2e. But it also recognizes that current earnings are reduced by investment that adds to future earnings but investment that conveys risk to these earnings.

These results support the critique of the FF comparative static properties (i) and (ii), but there remains the prediction on investment in property (iii) in the FF. Though not implied by that modelling, the FF use $\Delta \text{Total Assets}_t / \text{Total Assets}_{t-1}$ as the measure of investment. So Regression VI adds this measure and, as in FF, it returns a negative coefficient. However, balance sheet investment involves applying cash to purchase assets with no effect on $\Delta \text{Total Assets}_t$, nor on book value in property (iii). Also, $\Delta \text{Total Assets}_t$ include investment in cash (often in low-risk cash and Treasuries) with lower risk but not the risk of operations. So, to capture effective investment in the operating business, Regression VII defines investment as the change in net operating assets (NOA) in the business, $\Delta \text{NOA}_t / P_{t-1}$. The mean rank correlation between this investment measure and $\Delta \text{Total Assets}_t / \text{Total Assets}_{t-1}$ is 0.615. The denomination by price in $\Delta \text{NOA}_t / P_{t-1}$ is the same denomination as that for ExpInv , E/P , and forward returns, though with a lag. The coefficient also is reliably negative.¹⁷ When $\Delta \text{Total Assets}_t / \text{Total Assets}_{t-1}$ is added in Regression VIII, its coefficient estimate is close to zero.

These are just correlations but the negative relation between investment and returns observed from data mining and (conditionally) here is predicted by an accounting principle, the realization principle under which earnings are booked only when risk is largely resolved: The firm has won a customer and receipt of cash is “highly certain.” Investment booked to the balance sheet, in contrast to that charged against earnings, projects the ability to realize earnings in the future and thus risk reduction. That resonates with the common view in finance that investment is the exercise of risky growth options, risk resolved.

The final Regression IX in Table 2 adds Beta, Size, and Momentum, variables in the FF factor models, including those in the six-factor model. None of these variables return a significant coefficient though there is an increase in R^2 .

informs that expensed investment is of higher risk than that booked to the balance sheet. Bongaerts, Kang, and van Dijk (2021) calculate an intangible asset factor based on expensed investment.

¹⁷ That is so for other measures of investment used in the literature, including capital expenditure.

Returns from Portfolio Sorts

As Fama and French (2020) explain, estimated slope coefficients in these regressions are estimates of factor returns, with asset characteristics then the sensitivity of returns to the factors. In the spirit of taking characteristic regression results to factor construction, Tables 3 and 4 report mean annual returns for “factor mimicking” based on the characteristics. Portfolios are formed each March 31 from rankings on the respective accounting numbers from firms’ most recent annual reports. For December 31 fiscal-year firms, the majority, the accounting information is most immediate. As accounting numbers are codetermined in the double-entry system, conveying information jointly, the portfolio formation is with nested sorts, not the typical independent sorts as if the numbers were separate, independent factors that explain returns.

Table 3 follows the FF modelling, reporting mean equally weighted annual returns from 5 x 5 sorts on B/P and Operating Profitability (OP), the profitability measure they choose based on its performance in goodness-of-fit tests rather than ROE in the comparative statics. With this modification, these are the two measures in FF properties (i) and (ii), with the portfolio construction varying OP while holding B/P constant as in property (ii).

In Panel A of Table 3, the correlation of portfolios returns with B/P is clear, the basis for a B/P factor. For a given B/P, there is not much incremental return from variation on OP, as the t-statistics at the bottom of the panel indicate. However, the difference between the Low-Low portfolio return of 8.0% and the 16.2% for the High-High portfolio is significant with a t-statistic of 2.03 (reported in the bottom right-hand corner of the table). FF form these portfolios within a size split and the return spread is higher for small firms than large firms in Panels B and C. It is preserved with the 3-way split in Panel D and, though diminished, with value weighted returns in Panel E.

For a given B/P in Panel A of Table 3, the sort on Operating Profitability is a sort on a quasi-E/P ratio:

$$B_t/P_t \times OP_t/B_{t-1} = OP_t/P_t$$

(approximately).¹⁸ Thus it is recovering a varying E/P ratio (with operating profits as earnings) for a given B/P. But operating profits are not quite the same as net income in E/P if it omits R&D. This seems to be so on the Kenneth French website though the Fama and French (2018) calculation appears to include R&D.¹⁹ If OP omits R&D, that is interesting because the second sort in Table 3 yields a quasi-E/P that adjusts earnings for expensed investment.

Table 4 carries out a similar 5 x 5 portfolios analysis that explicitly incorporates the expensed investment and the risk it conveys. Portfolios are first formed from ranking on positive E_t/P_t , the product of E/B and B/P as in equation 3. Then, within each E_t/P_t portfolio, portfolios are formed from ranking on ExpInv_t/P_t . The panels in the table correspond to those in Table 3.

It is clear from the All E/P row in Panel A of Table 4 that E/P is positively related to returns. The return difference of 6.0% between high and low E/P is less than the 12.5% for B/P in Table 3, perhaps prompting B/P rather than E/P as a basis for a factor.²⁰ However, given E/P, ExpInv_t/P_t adds to the discrimination of returns with higher ExpInv_t/P_t conveying higher returns for risk. In contrast to Table 3, the return differences down columns are highly significant, and that is so for both small and large firms in Panels B and C except for high E/P for large firms. The spread of returns from Low-Low to High-High in Panel A, the mean return from a potential long-short zero-net-investment factor, is 18.0% with a t-statistic of 3.94. The mean return is 13.9% ($t = 3.73$) with the 3-way split and 7.0% ($t = 2.02$) with value weighted returns that weight away from small firms. If these mean returns are estimates of expected returns, E/P adjusted for the expensed investment that conveys risk, indicates expected returns.

To incorporate investment in property (iii), Panel A of Table 5 adds five portfolios from a sort on the FF investment measure, $\Delta \text{Total Assets}_t / \text{Total Assets}_{t-1}$ within the High E_t/P_t , High ExpInv_t/P_t portfolio and the Low E_t/P_t , Low ExpInv_t/P_t portfolio in Panel A of Table 4. Panel B

¹⁸ While Operating Profitability is defined on a base of lagged book value on the Kenneth French website, it is on end-of-period book value (at t) for the construction of the RMW factor in Fama and French (2018), in which case the “approximately” qualification can be removed.

¹⁹ The confusion could arise from the Compustat calculation of SG&A that includes R&D.

²⁰ In equation 2e, expected future earnings-to-price that yields r must be positive for a positive price. So the ranking in Table 4 is with positive earnings-to-price that projects future earnings-to-price. While returns are monotonically increasing in E/P in Panel A of Table 4, they are slightly U-shaped when loss firms included, with mean returns for negative E/P higher than those for the lowest positive E/P portfolio.

repeats but with the $\Delta NOA_t/P_{t-1}$ as the investment measure, and Panels C and D with 3-way splits.

In all panels, higher investment is associated with lower returns for both the High E_t/P_t , High $ExpInv_t/P_t$ portfolio and the Low E_t/P_t , Low $ExpInv_t/P_t$. The High minus Low difference for Low E_t/P_t , Low $ExpInv_t/P$ (in the right lower corner) is statistically significant in all panels, and there does not appear to be a large difference between the two alternative investment measures that are mean rank correlated 0.615. We emphasize that the portfolio construction is not from data mining, nor from the FF comparative statics, but from an understanding of how accounting numbers convey information about risk and return.

Robustness Over Time

Table 6 compares potential factor mimicking portfolios in subperiods. These are the zero-investment portfolios with 3-way splits in the lower right corner of panels D in Tables 3, 4, and 5. The $E^+/P_ExpInv_ \Delta NOA$ split dominates in all subperiods, both on mean returns and (annual) Sharpe Ratios.

Figure 1 plots the cumulative annual returns from 1964 to 2022 for the three portfolios. The higher return for $E^+/P_ExpInv_ \Delta NOA$ is clear and that for the 3-way split on E^+/P and $ExpInv/P$ is higher than that for B/P and Operating Profitability. The returns accumulate over all subperiods.

Summary and Conclusion

With a model that explains stock prices with accounting information, Fama and French (2015) carry out some comparative statistics that are said to identify accounting characteristics, book-to-price (B/P), profitability, and investment from which pricing factors are then constructed. The modelling is much needed given relevant characteristics are often identified simply by data mining (without theory). However, while the model is consistent with pricing theory, the comparative statics applied to identify characteristics are inconsistent with the way a double-entry accounting system works. In that system, numbers are jointly determined (each debit has a corresponding credit) so that one cannot hold one feature constant while varying another.

Further, the accounting numbers in the pricing model are identified by substituting earnings and book value for the dividends in the standard infinite-horizon dividend discount model. While

this yields an equivalent price to that for discounting dividends, mere substitution puts no information on the table. Further conditions on the measurement of the accounting numbers under which they convey the risk imputed to them are required; indeed, for finite forecast horizons the earnings and book value in the model can be random numbers with the equivalence to the price from the dividend model enforced only with a transversality condition.

The Alternative Accounting Logic

With the same accounting-based model, the paper lays out an alternative identification of characteristics based on the accounting principles that govern that system. The model implies that the expected return is also given by expected future earnings relative to current price, that is, the expected earnings yield, so that information that informs about that yield informs about the expected return. The current earnings yield, denominated in the same price, serves to the extent that it forecasts the future yield. The yield is implicitly in the B/P and profitability measures in the Fama and French analysis. However, rather than B/P varying while holding profitability constant and vice versa under their comparative statics, the two are jointly determined and combine to the current yield by the calculation, $B/P \times E/B = E/P$ for all B and E.

Current E/P is typically not a sufficient statistic for the expected earnings yield for earnings are disturbed by expensing investment that gains earnings. However, under accounting principles, that accounting conveys risk. Accordingly, E/P along with expensed investment relative to price conveys the expected earnings yield. The accounting also identifies investment booked to the balance sheet as conveying risk as in the Fama and French identification but with an alternative investment measure to the Fama and French measure.

The empirical analysis supports the critique and the alternative characteristic identification and resulting factor construction although the alternative investment measure versus the Fama and French measure does not make much difference in discriminating amongst returns. The paper only focuses on the three attributes in the Fama and French modelling, not a thorough factor identification. Other accounting features may well enter into forecasting the expected earnings yield. Indeed, Penman and Zhu (2022) and Penman and Zhang (2024), following accounting principles, add other accounting numbers to factor models which, in standard spanning tests, perform well.

Appendix. Calculation of Variables

This appendix details the calculation of variables used in the empirical analysis. Compustat sometimes reports an item as missing. These are either in the financial reports as zero, aggregated with other items in the financial reports, or reported but not found by Compustat. We follow the practice of setting these to zero, relying on the finding in Casey, Gao, Kirschenheiter, Li, and Pandit (2016) that setting to zero satisfies certain articulating accounting equations 82.7% of the time. Nevertheless, the remaining 17.3% are a qualification to our analysis.

Ret_{t+1}	One-year, buy-and-hold return calculated as compounded CRSP monthly returns, starting at the beginning of the fourth month after the current fiscal year end. For firms that are delisted during the 12 months, the return for the remaining months is calculated by first applying the CRSP delisting return and then reinvesting any remaining proceeds at the risk-free rate. This mitigates concerns with potential survivorship biases. Firms that are delisted for poor performance (delisting codes 500 and 520-584) frequently have missing delisting returns (Shumway 1997). We control for this potential bias by applying delisting returns of -55% for NASDAQ firms and -30% for NYSE/AMEX firms (Shumway and Warthar 1999). Results are qualitatively similar if we make no such adjustment, or assuming a uniform -100% return in such cases. Accounting data for a fiscal year are presumed to have been published during the three months after fiscal-year end, as required by law (and before the beginning of the return period).
$Earnings_t$	Earnings before extraordinary items available for common (Compustat item IBCOM) and special items (Compustat item SPI), with a tax allocation to special items at the prevailing federal statutory corporate income tax rate for the year. The top statutory federal tax rate was 50% in 1964, 48% in 1965-1967, 52.8% in 1968-1969, 49.2% in 1970, 48% in 1971-1978, 46% in 1979-1986, 40% in 1987, 34% in 1988-1992, 35% in 1993-2017, and 21% in 2018.
B_t	Book value of common equity at the end of the fiscal-year, calculated as Compustat common equity (Compustat item CEQ) plus any preferred treasury stock (Compustat item TSTKP) less any preferred dividends in arrears (Compustat item DVPA).
P_t	Market value of equity three months after fiscal-year end. It is calculated as the number of shares outstanding multiplied by the price per share from CRSP at three months after fiscal-year end. It can also be calculated as the number of shares outstanding at the end of the fiscal year from Compustat (#CSHO) multiplied by the price per share from CRSP at three months after fiscal-year end, adjusted for any intervening stock splits and stock dividends. This excludes any change in the market price from net share issues over the three months when calculating E/P and B/P ratios based on earnings and book value at year end. Market price can also be calculated as per-share price at three months after fiscal-year end, adjusted for stock splits and stock dividends over the three months, in which case, earnings and book value are on a per-share basis. The first calculation is applied in this paper as that seems to be the practice for E/P and B/P calculations in asset pricing papers to which this paper compares.
TA_t	Total assets (Compustat item AT).

Investment _t (FF)	Change in Total Assets. This is the measure in Fama and French (2015 and 2018) five- and six-factor models.
ExpInv _t	$\text{ExpInv}_t = \text{R\&D}_t + \text{Advertising}_t + \text{SG\&A}_t \times 0.3$. R&D is Compustat item XRD, Advertising is Compustat item XAD and the SG&A variable is $\text{XSGA} - \text{XAD} - \text{XRD}$ when XSGA is greater than the sum of XAD and XRD, otherwise SG&A equals XSGA. Including 1/3 of SG&A recognizes that a significant portion of expensed investment is charged to that account, as estimated in Peters and Taylor (2017). Results are similar with ExpInv _t calculated without the SG&A component.
NOA _t	Net operating assets at the end of the fiscal year, measured as Net Debt plus book value of common equity, B _t , with added non-controlling (minority) interest (Compustat item MIB).
ROE _t	$\text{Earnings}_t / \text{B}_{t-1}$, with Earnings _t calculated as above.
OP _t	Operating profits defined as revenue minus cost of goods sold minus SG&A minus interest on the Kenneth French website, though it appears to include R&D in Fama and French (2018). On the website OP _t measure in Fama and French is deflated by lagged book value but by current book value (at t) in Fama and French (2018).
Beta _t	Stock market betas estimated from monthly stock returns during the previous five years ending in the third month after the current fiscal year end, or a lessor period if five years of data are not available.
Size _t	Logarithm of market value of equity _t .
MOM _t	Twelve month buy-and-hold return ending in the third month after the current fiscal year end.

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Figure 1. Cumulative Annual Returns for Factor Portfolios, 1964-2022.

BP_OP is the return on the (High, High) portfolio minus that on the (Low, Low) portfolio in Panel D of Table 3 from the spread on book-to-price (B/P) and the Fama and French Operating Profitability measure (OP) with a 3-way split.

E⁺/P_ExpInv is the return on the (High, High) portfolio minus that on the (Low, Low) portfolio in Panel D of Table 4 from the spread on E⁺/P and expensed investment, ExpInv/P with a 3-way split.

E⁺/P_ExpInv_dNOA is the return on the (High, High, Low) portfolio minus that on the (Low, Low, High) portfolio in Panel A of Table 5 from the spread on E⁺/P and expensed investment, ExpInv/P, and balance sheet investment measured as $\Delta\text{NOA}_t/P_{t-1}$.

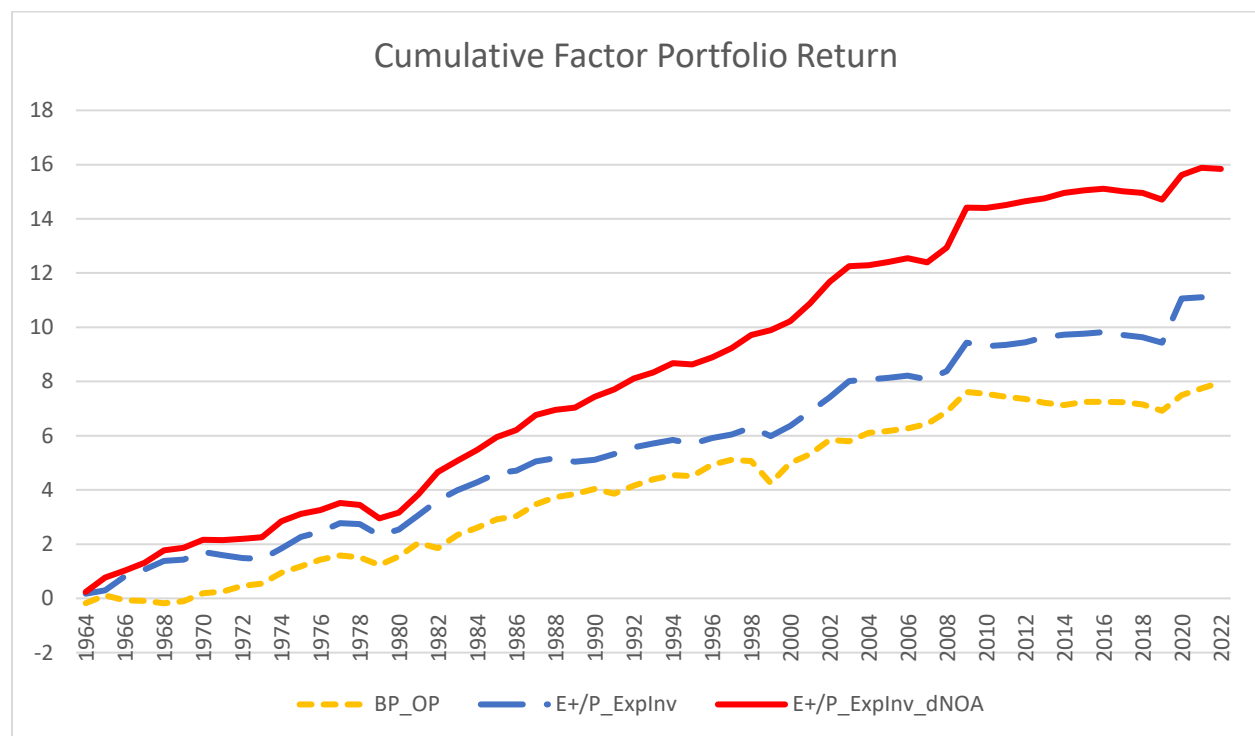


TABLE 1**Mean Return on Equity (ROE) and Earnings-to-Price (E/P) for Book-to-Price (B/P) Portfolios**

Panel A reports the mean book-to-price (B/P) and return on equity (ROE) over years, 1964-2022, for portfolios formed on annual sorts on B/P into five portfolios. The first ROE row is for firms with positive ROE (ROE⁺), the second including loss firms. Firms with negative lagged book values are excluded. Panel B reports mean trailing E/P ratios for the same B/P portfolios for positive E⁺/P firms and all E/P including loss firms. Panel C reports mean E/P for E⁺/P portfolios formed each year within five portfolios formed on B/P, and Panel D for all E/P. Firms with negative book values are excluded in both panels. See the data section for the sample coverage.

Panel A: ROE for B/P Portfolios

B/P Portfolios	Low	2	3	4	High
B/P	0.191	0.407	0.591	0.838	1.699
ROE ⁺	0.351	0.214	0.170	0.134	0.095
All ROE	-0.061	0.059	0.054	0.036	-0.023

Panel B: E/P for B/P Portfolios

B/P Portfolios	Low	2	3	4	High
B/P	0.191	0.407	0.591	0.838	1.699
E ⁺ /P	0.046	0.064	0.078	0.089	0.115
All E/P	-0.018	0.015	0.021	0.015	-0.090

Panel C: E/P for E⁺/P Portfolios Formed Within B/P Portfolios

	B/P	1 (Low)	2	3	4	5 (High)
E ⁺ /P	1 (Low)	0.013	0.026	0.030	0.032	0.028
	2	0.030	0.049	0.058	0.065	0.067
	3	0.042	0.062	0.074	0.084	0.098
	4	0.054	0.075	0.090	0.105	0.136
	5 (High)	0.088	0.109	0.134	0.159	0.245

Panel D: All E/P (Including Loss Firms) for E/P Portfolios Formed Within B/P Portfolios

	B/P	1 (Low)	2	3	4	5 (High)
E/P	1 (Low)	-0.221	-0.017	-0.202	-0.265	-0.730
	2	-0.016	0.023	0.030	0.024	-0.071
	3	0.024	0.053	0.064	0.069	0.044
	4	0.044	0.069	0.084	0.095	0.099
	5 (High)	0.078	0.103	0.126	0.148	0.204

TABLE 2

The table reports mean coefficient estimates from annual cross-sectional decile-rank regressions of annual forward stock returns, Ret_{t+1} on accounting numbers on the left-hand side of the table, 1964-2022. t-statistics (in parentheses) are calculated as the mean coefficients relative to their standard errors estimated from the time-series of coefficient estimates. Firms with negative lagged book values are excluded. Variables are defined in the appendix.

	I	II	III	IV	V	VI	VII	VIII	IX
Intercept	0.071 (2.20)	0.056 (1.71)	0.103 (3.60)	0.130 (4.30)	0.076 (2.92)	0.108 (4.49)	0.119 (4.73)	0.118 (4.94)	0.141 (3.21)
B_t/P_t	0.020 (4.26)	0.018 (3.84)							0.002 (1.02)
ROE_t^+		0.006 (2.45)		-0.008 (-3.38)	-0.003 (-1.46)				
E_t^+/P_t			0.014 (3.97)	0.016 (4.06)	0.010 (3.11)	0.009 (3.05)	0.009 (3.36)	0.010 (3.46)	0.006 (2.93)
$ExpInv_t/P_t$					0.013 (4.19)	0.013 (3.80)	0.014 (4.34)	0.014 (4.21)	0.012 (4.80)
$\Delta TA_t/TA_{t-1}$						-0.007 (-3.30)		-0.000 (-0.04)	
$\Delta NOA_t/P_{t-1}$							-0.011 (-5.54)	-0.011 (-5.40)	-0.011 (-5.43)
$Beta_t$									0.000 (0.10)
$Size_t$									0.001 (0.43)
MOM_t									-0.004 (-1.02)
Adjusted R^2	0.015	0.022	0.016	0.023	0.032	0.033	0.034	0.037	0.068

TABLE 3

Panels A-C report mean equally weighted annual returns for 5 x 5 FF portfolios from sorts on B/P and operating profitability (OP) at March 31 each year, 1964-2022, the two measures from properties (i) and (ii) in the FF comparative statics. Small (large) firms are those below (above) the NYSE median market capitalization each year. Panel D repeats the analysis with a three-way sort more common in factor construction, with high (low) portfolios composed of the top (bottom) 30% of stocks from the ranking. Panel E repeats the analysis in Panel D but with value weighted returns. The High-Low return in the lower right-hand corner of each panel is the difference in mean returns between the (High, High) portfolio and the (Low, Low) portfolio. t-statistics are mean differences in High and Low returns relative their standard errors estimated from the time series of means. Variables are defined in the appendix.

Panel A: Returns for FF Portfolios Formed on B/P and Operating Profitability (all firms)

	B/P					High – Low (t-stat)
	1 (Low)	2	3	4	5 (High)	
All	0.106	0.140	0.160	0.196	0.228	0.122 (3.47)
1 (Low)	0.080	0.118	0.154	0.200	0.232	0.152 (4.14)
2	0.149	0.166	0.153	0.194	0.227	0.078 (2.32)
OP 3	0.110	0.145	0.158	0.192	0.236	0.127 (2.93)
4	0.128	0.153	0.171	0.192	0.208	0.080 (2.16)
5 (High)	0.131	0.147	0.171	0.165	0.162	0.033 (0.85)
High-Low (t-stat)	0.051 (1.66)	0.030 (0.86)	0.017 (0.46)	-0.036 (-0.93)	-0.070 (-2.08)	0.082 (2.03)

Panel B: Returns for FF Portfolios Formed on B/P and Operating Profitability (small firms)

	B/P					
	1 (Low)	2	3	4	5 (High)	High – Low (t-stat)
All	0.110	0.151	0.173	0.208	0.235	0.125 (3.74)
1 (Low)	0.077	0.125	0.166	0.206	0.239	0.162 (4.40)
2	0.177	0.204	0.173	0.206	0.235	0.057 (1.34)
3	0.122	0.168	0.169	0.206	0.243	0.121 (2.99)
4	0.145	0.179	0.181	0.203	0.217	0.072 (1.71)
5 (High)	0.140	0.152	0.187	0.174	0.166	0.026 (0.96)
High-Low (t-stat)	0.062 (2.25)	0.027 (1.11)	0.020 (0.50)	-0.033 (-0.85)	-0.073 (-2.01)	0.090 (2.03)

Panel C: Returns for FF Portfolios Formed on B/P and Operating Profitability (large firms)

	B/P					
	1 (Low)	2	3	4	5 (High)	High – Low (t-stat)
All	0.126	0.130	0.135	0.150	0.165	0.040 (1.43)
1 (Low)	0.125	0.099	0.093	0.129	0.159	0.035 (0.68)
2	0.147	0.120	0.128	0.163	0.174	0.027 (0.80)
3	0.118	0.129	0.147	0.154	0.165	0.048 (1.66)
4	0.120	0.134	0.148	0.152	0.135	0.015 (0.42)
5 (High)	0.132	0.141	0.130	0.118	0.131	0.002 (0.05)
High-Low (t-stat)	0.007 (0.07)	0.042 (1.67)	0.038 (1.73)	-0.012 (-0.45)	-0.029 (-0.58)	0.006 (0.10)

Panel D: Returns for FF Portfolios Formed on B/P and Operating Profitability with 3-way splits (all firms, equally weighted returns)

	B/P			
	1 (Low)	2	3 (High)	High – Low (t-stat)
	All	0.111	0.166	0.219
				0.108 (3.52)
OP	1 (Low)	0.087	0.167	0.223
				0.135 (4.30)
	2	0.129	0.164	0.219
				0.090 (2.85)
	3 (High)	0.134	0.166	0.183
				0.050 (1.81)
	High-Low (t-stat)	0.046 (1.65)	-0.001 (-0.06)	-0.039 (-1.88)
				0.096 (3.09)

Panel E: Returns for FF Portfolios Formed on B/P and Operating Profitability with 3-way splits (all firms, value weighted returns)

	B/P			
	1 (Low)	2	3 (High)	High – Low (t-stat)
	All	0.116	0.116	0.151
				0.035 (1.50)
OP	1 (Low)	0.097	0.120	0.148
				0.052 (1.69)
	2	0.115	0.120	0.155
				0.040 (1.98)
	3 (High)	0.121	0.114	0.151
				0.030 (0.95)
	High-Low (t-stat)	0.024 (0.89)	-0.006 (-0.34)	0.002 (0.10)
				0.054 (1.58)

TABLE 4

Panels A-C report mean equally weighted annual returns for 5 x 5 portfolios from sorting first on positive E_t/P_t at March 31 each year, 1964-2022, then, within E_t/P_t portfolios, sorting on expensed investment, ExpInv_t/P_t . Small (large) firms are those below (above) the NYSE median market capitalization each year. Panel D repeats the analysis with a three-way sort with high (low) portfolios composed of the top (bottom) 30% of stocks from the ranking. Panel E repeats the analysis in Panel D but with value weighted returns. The High-Low return in the lower right-hand corner of each panel is the difference in mean returns between the (High, High) portfolio and the (Low, Low) portfolio. t-statistics are mean differences in High and Low returns relative to their standard errors estimated from the time series of means. Variables are defined in the appendix.

Panel A: Returns for Portfolios Formed on E_t/P_t and ExpInv_t/P_t (all firms)

		E^+/P					
		1 (Low)	2	3	4	5 (High)	High – Low (t-stat)
ExpInv	All	0.142	0.151	0.166	0.180	0.201	0.060 (2.55)
	1 (Low)	0.081	0.117	0.109	0.134	0.152	0.071 (2.45)
	2	0.104	0.132	0.146	0.161	0.185	0.081 (2.72)
	3	0.132	0.144	0.171	0.176	0.193	0.062 (2.54)
	4	0.163	0.159	0.178	0.198	0.215	0.052 (1.62)
	5 (High)	0.228	0.201	0.225	0.232	0.261	0.033 (1.36)
	High-Low (t-stat)	0.147 (3.61)	0.085 (2.72)	0.116 (3.80)	0.098 (3.55)	0.109 (3.82)	0.180 (3.94)

Panel B: Returns for Portfolios Formed on E_t/P_t and $ExpInv_t/P_t$ (small firms)

		E^+/P					
		1 (Low)	2	3	4	5 (High)	High – Low (t-stat)
ExpInv	All	0.165	0.174	0.182	0.191	0.210	0.045 (2.00)
	1 (Low)	0.081	0.123	0.125	0.131	0.156	0.075 (2.28)
	2	0.099	0.145	0.153	0.169	0.186	0.088 (2.95)
	3	0.153	0.152	0.187	0.189	0.200	0.047 (1.73)
	4	0.164	0.188	0.186	0.201	0.222	0.058 (1.69)
	5 (High)	0.239	0.216	0.238	0.242	0.265	0.026 (1.00)
	High-Low (t-stat)	0.158 (3.41)	0.093 (2.40)	0.112 (3.35)	0.111 (3.58)	0.109 (4.02)	0.184 (3.98)

Panel C: Returns for Portfolios Formed on E_t/P_t and $ExpInv_t/P_t$ (large firms)

		E^+/P					
		1 (Low)	2	3	4	5 (High)	High – Low (t-stat)
ExpInv	All E^+/P	0.122	0.126	0.141	0.158	0.158	0.035 (1.39)
	1 (Low)	0.091	0.112	0.105	0.142	0.143	0.052 (1.71)
	2	0.123	0.123	0.140	0.151	0.167	0.044 (1.35)
	3	0.129	0.135	0.151	0.149	0.164	0.035 (1.31)
	4	0.144	0.128	0.158	0.182	0.157	0.014 (0.45)
	5 (High)	0.165	0.157	0.175	0.180	0.175	0.010 (0.27)
	High-Low (t-stat)	0.074 (2.23)	0.045 (2.26)	0.070 (3.27)	0.038 (1.82)	0.031 (0.93)	0.084 (1.84)

Panel D: Returns for Portfolios Formed on E_t/P_t and $ExpInv_t/P_t$ with 3-way splits (all firms, equally weighted returns)

		E^+/P			
		1 (Low)	2	3 (High)	High – Low (t-stat)
ExpInv	All	0.145	0.165	0.195	0.050 (2.63)
	1 (Low)	0.097	0.126	0.156	0.058 (2.52)
	2	0.133	0.163	0.194	0.061 (2.86)
	3 (High)	0.204	0.205	0.236	0.032 (1.75)
	High-Low (t-stat)	0.107 (3.52)	0.079 (4.16)	0.080 (3.67)	0.139 (3.73)

Panel E: Returns for Portfolios Formed on E_t/P_t and $ExpInv_t/P_t$ with 3-way splits (all firms, value weighted returns)

		E^+/P			
		1 (Low)	2	3 (High)	High – Low (t-stat)
ExpInv	All	0.111	0.124	0.145	0.035 (1.72)
	1 (Low)	0.104	0.110	0.130	0.026 (1.08)
	2	0.120	0.133	0.164	0.044 (1.97)
	3 (High)	0.140	0.165	0.175	0.034 (1.11)
	High-Low (t-stat)	0.036 (1.78)	0.055 (4.07)	0.044 (1.50)	0.070 (2.02)

TABLE 5

The table reports mean returns for five portfolios formed March 31 of each year, 1964-2022 from a ranking of firms on investment within the (High E_t/P_t , High ExpInv_t/P_t) portfolio and the (Low E_t/P_t , Low ExpInv_t/P_t) portfolio in Panel A of Table 4. The investment measure in Panel A is $\Delta\text{Total Assets}_t/\text{Total Assets}_{t-1}$ and that in Panel B is $\Delta\text{NOA}_t/P_{t-1}$. Panel C repeats the analysis in Panel B but with three investment portfolios. The High-Low return in the lower right-hand corner of each panel is the difference in mean returns between the (High, High, Low) portfolio and the (Low, Low, High) portfolio. t-statistics are mean differences in High and Low returns relative its standard error estimated from the time series of means. Variables are defined in the appendix.

Panel A: Investment Measure is $\Delta\text{Total Assets}_t/\text{Total Assets}_{t-1}$

	1 (Low)	2	3	4	5 (High)	High – Low
Low E_t/P_t , Low ExpInv_t/P_t	0.114	0.107	0.098	0.108	0.070	-0.045 (-1.99)
High E_t/P_t , High ExpInv_t/P_t	0.301	0.265	0.211	0.223	0.195	-0.106 (-3.34)
High, High, Low – Low, Low, High (t-statistic)	0.187 (3.64)	0.159 (3.95)	0.113 (3.13)	0.115 (3.06)	0.126 (3.11)	0.231 (4.80)

Panel B: Investment Measure is $\Delta\text{NOA}_t/P_{t-1}$

	1 (Low)	2	3	4	5 (High)	High – Low
Low E_t/P_t , Low ExpInv_t/P_t	0.130	0.136	0.090	0.079	0.077	-0.053 (-2.63)
High E_t/P_t , High ExpInv_t/P_t	0.321	0.251	0.219	0.212	0.187	-0.134 (-3.82)
High, High, Low – Low, Low, High (t-statistic)	0.191 (3.72)	0.115 (2.92)	0.129 (2.69)	0.133 (3.60)	0.110 (2.61)	0.244 (5.14)

Panel C: Portfolios with 3-way Splits for $\Delta \text{Total Assets}_t / \text{Total Assets}_{t-1}$

	1 (Low)	2	3 (High)	High – Low
High E_t/P_t , High ExpInv_t/P_t	0.109	0.106	0.082	-0.027 (-1.41)
Low E_t/P_t , Low ExpInv_t/P_t	0.289	0.221	0.206	-0.084 (-3.40)
High, High, Low – Low, Low, High (t-statistic)	0.180 (3.95)	0.115 (3.04)	0.123 (3.37)	0.207 (4.73)

Panel D: Portfolios with 3-way Splits for $\Delta \text{NOA}_t / P_{t-1}$

	1 (Low)	2	3 (High)	High – Low
High E_t/P_t , High ExpInv_t/P_t	0.130	0.102	0.075	-0.055 (-2.80)
Low E_t/P_t , Low ExpInv_t/P_t	0.290	0.226	0.197	-0.093 (-4.15)
High, High – Low, Low (t-statistic)	0.160 (3.55)	0.124 (3.00)	0.122 (3.35)	0.215 (5.44)

TABLE 6

Subperiod results for potential factor mimicking portfolios. These are portfolios in the lower right-hand corner of panels D in Tables 3, 4, and 5 with 3-way splits. BP_OP is the return on the (High, High) portfolio minus that on the (Low, Low) portfolio in Panel D of Table 3 from the spread on book-to-price (B/P) and the Fama and French Operating Profitability measure (OP) with a 3-way split. E⁺/P_ExpInv is the return on the (High, High) portfolio minus that on the (Low, Low) portfolio in Panel D of Table 4 from the spread on E⁺/P and expensed investment, ExpInv/P with a 3-way split. E⁺/P_ExpInv_dNOA is the return on the (High, High, Low) portfolio minus that on the (Low, Low, High) portfolio in Panel A of Table 5 from the spread on E⁺/P, expensed investment, ExpInv/P, and balance sheet investment measured as $\Delta\text{NOA}_t/P_t$.

1.

Panel A: All Years:

Portfolios	Mean	Std	Sharpe Ratio
B/P_OP	0.096	0.239	0.215
E ⁺ /P_ExpInv	0.139	0.286	0.322
E ⁺ /P_ExpInv_dNOA	0.215	0.303	0.549

Panel B: 1964-1983

Portfolios	Mean	Std	Sharpe Ratio
B/P_OP	0.056	0.209	-0.064
E ⁺ /P_ExpInv	0.119	0.214	0.232
E ⁺ /P_ExpInv_dNOA	0.156	0.207	0.413

Panel C: 1984-2003

Portfolios	Mean	Std	Sharpe Ratio
B/P_OP	0.148	0.293	0.331
E ⁺ /P_ExpInv	0.157	0.193	0.521
E ⁺ /P_ExpInv_dNOA	0.296	0.221	1.057

Panel D: 2004-2022

Portfolios	Mean	Std	Sharpe Ratio
B/P_OP	0.084	0.206	0.347
E ⁺ /P_ExpInv	0.140	0.418	0.304
E ⁺ /P_ExpInv_dNOA	0.192	0.434	0.411